Temperature Extension Module for BMS – Design Report

Abstract

This document presents the design and implementation approach for a Temperature Extension Module intended to integrate with an existing Battery Management System (BMS) architecture based on boards sponsored by Analog Devices. **The primary objective of the module is to acquire temperature data from multiple sensors distributed across the battery pack and deliver the data to the BMS master controller within a <1s acquisition window**. The design focuses on improving signal integrity, reducing electromagnetic interference (EMI), minimizing wiring complexity, and adopting robust automotive-style communication methods suitable for electrically noisy battery environments.

1. Design Goals

- Ensure compatibility with the existing BMS architecture utilizing Analog Devices evaluation and support boards.
- Enable high-speed temperature acquisition from multiple sensing points within a 1-second sampling window.
- Minimize susceptibility to EMI/EMC in the battery pack environment.
- Reduce analog signal routing distance and associated noise coupling.
- Adopt an industry-standard and automotive-grade communication protocol.
- Maintain scalability for additional sensor channels in future system expansions.

2. Early System Architecture and Key Learnings

The initial prototype architecture relied on an **analog multiplexing** approach using analog multiplexer ICs combined with external counters and clock sources. Temperature sensor signals were multiplexed and routed toward a centralized measurement stage.

- The multiplexed architecture limited achievable sampling speed due to sequential channel switching.
- Analog signals traversed long PCB traces and wiring paths within a high-current battery environment, resulting in noise susceptibility.
- The architecture was inefficient for integration with the CAN-based charger communication framework.
- The system complexity increased due to synchronization requirements between counters, clocks, and multiplexers.

3. Design Strategy

To address the limitations observed in the earlier system, the revised architecture focuses on digitizing temperature signals at the earliest possible stage.

By converting analog signals into digital data locally, signal integrity can be significantly improved while reducing the vulnerability to noise in the battery environment.

Furthermore, the design adopts the **Controller Area Network (CAN)** protocol, a well-established automotive communication standard known for its robustness, deterministic communication behavior, and strong noise immunity.

4. ADC Integration Considerations

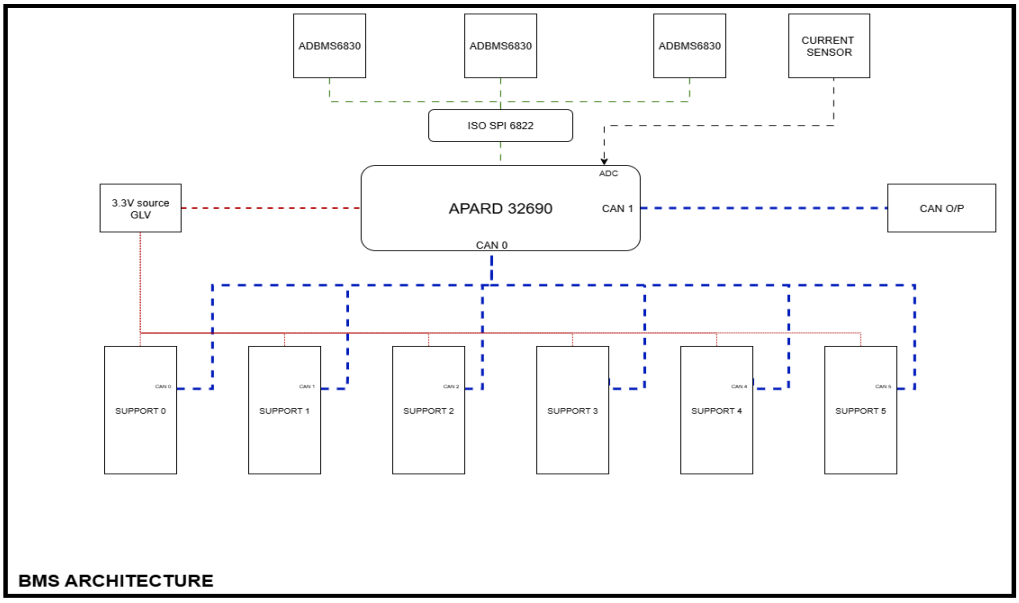
Temperature sensors produce analog signals that must be digitized using Analog-to-Digital Converters (ADCs). Typical ADC interfaces rely on protocols such as SPI or I2C. However, implementing discrete ADC devices with these interfaces introduces several design challenges:

- Each temperature board would require a dedicated communication channel.
- Synchronization across multiple ADC devices becomes difficult.
- The number of ADC channels required exceeds those available on many microcontrollers.
- SPI/I2C communication lines may become more susceptible to EMI in the battery environment.

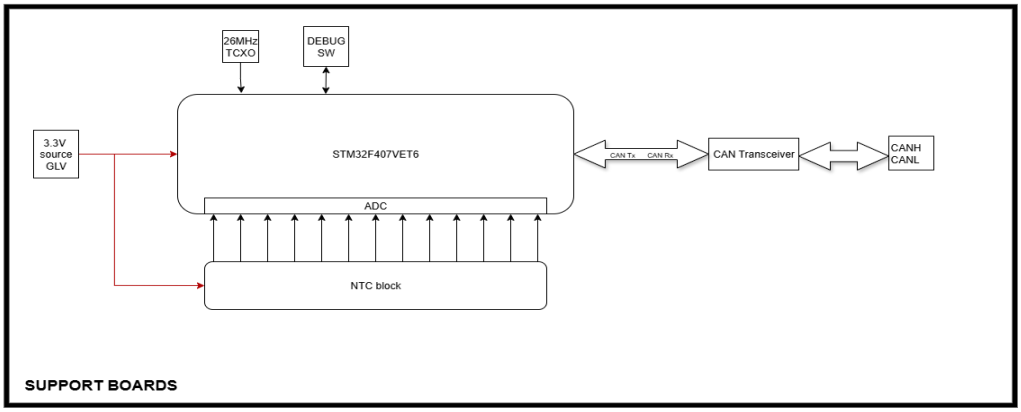
Microcontroller-Based Architecture

Considering the above constraints, the design decision was made to utilize distributed microcontroller-based nodes that perform local ADC acquisition and communicate with the central BMS master via the CAN bus. This architecture reduces analog signal routing, improves synchronization, and reduces wiring complexity.

The selected microcontroller communicates with the master controller (APARD 32690 platform) over the CAN network.



BMS ARCHITECTURE



SUPPORT BOARDS

System Architecture Overview

The temperature extension system is designed to integrate with the existing Battery Management System (BMS) architecture based on the ADBMS6830 and the APARD32690 master controller.

The architecture follows a distributed sensing topology, where dedicated temperature support modules perform local signal acquisition and digitization. The processed temperature data is transmitted to the BMS master controller using a robust automotive communication network.

This approach reduces analog signal routing within the battery pack, improving measurement integrity and system reliability in electrically noisy environments.

Master BMS Controller

The central controller of the system is the **APARD32690**, which coordinates battery monitoring and system communication.

Its primary responsibilities include:

- Communication with cell monitoring ICs using isolated SPI (isoSPI)
- Acquisition of pack current measurements
- Collection and aggregation of temperature data from distributed sensing modules
- Transmission of processed BMS telemetry over the CAN network

Cell voltage monitoring is performed using the ADBMS6830 battery monitoring ICs. Communication between the ADBMS6830 devices and the master controller is implemented using isoSPI, which provides robust communication in high-voltage and high-noise environments typical of traction battery systems.

Temperature Support Module Network

To extend the available number of temperature measurement channels, six distributed temperature support modules are deployed across the battery pack.

Each support module performs the following functions:

- Acquisition of temperature data from multiple NTC thermistors
- Conversion of analog sensor signals into digital values
- Transmission of processed temperature data to the BMS master controller

Communication between the support modules and the master controller is implemented using the Controller Area Network.

Using CAN communication provides several advantages:

- Reduced analog signal routing across the battery pack
- Improved immunity to electromagnetic interference
- Simplified wiring harness architecture
- Scalable and modular system expansion

Support Module Hardware Architecture.

The microcontroller is responsible for:

- Sampling thermistor signals through the integrated Analog-to-Digital Converter (ADC)
- Performing signal conditioning and data processing
- Packaging temperature measurements into CAN communication frames
- Transmitting telemetry data to the BMS master controller

The MCU operates using an external crystal oscillator footprint, allowing flexibility in selecting the final clock source based on system timing requirements. This approach simplifies prototyping while allowing future upgrades to higher stability oscillators such as TCXO if required.

A dedicated debug interface is included to support firmware development, testing, and system diagnostics.

Temperature Measurement Subsystem

Temperature sensing is implemented using NTC thermistors arranged in resistive divider networks.

The thermistor voltage output varies with temperature and is sampled by the microcontroller ADC.

Performing analog-to-digital conversion locally on each support module ensures:

- Short analog routing paths
- Reduced susceptibility to electromagnetic interference
- Improved measurement accuracy in high-current battery environments

CAN Communication Interface

Communication between the temperature support modules and the BMS master controller is achieved using a dedicated CAN transceiver.

The transceiver performs physical-layer translation between:

- Microcontroller logic-level CAN signals (CAN_TX, CAN_RX)
- Differential CAN bus signals (CANH, CANL)

SELECTION OF KEY COMOPENTS

1. Microcontroller Selection

Several microcontroller families were evaluated for the design. Automotive-grade options such as NXP S32 series and STMicroelectronics AURIX-class devices were considered. However, due to project timeline constraints, limited development support, and budget considerations, a more accessible and well-supported development ecosystem was preferred.

STM32 microcontrollers were selected due to their mature development ecosystem, wide community support, familiarity within the development team, and strong peripheral set.

Within the STM32 family, three categories were considered:

- **Low-Power Series** – Suitable for energy-sensitive applications but limited by CAN availability and ADC performance.
- **High-Performance Series** – Provides advanced features such as higher resolution ADCs and CAN FD, but increases cost and exceeds system requirements.
- **Foundation (F-Series)** – Offers balanced performance, sufficient ADC capability, and integrated CAN support.

After evaluating multiple devices within the F-series, the STM32F407 was selected due to its strong ecosystem support, adequate ADC capability, high processing performance based on the ARM Cortex-M4 core, and availability in development tools and documentation.

The chosen device is STM32F407VET6 in a 100-pin LQFP package.

2. CAN Transceiver Selection

A dedicated CAN transceiver is required to interface the microcontroller CAN controller with the physical CAN bus. Several options were evaluated based on availability, robustness, and compatibility.

- **TJA1051** – Widely used and readily available CAN transceiver.
- **TCAN1044** – Modern device with 3.3 V compatibility but limited availability.
- **SN65 Series** – Mature solution but considered older generation.

Based on availability and industry adoption, **the TJA1051 CAN transceiver was selected for the design**.

3. Power Architecture

The system operates primarily from a 5 V supply rail, while the STM32 microcontroller requires a regulated 3.3 V supply. To minimize thermal dissipation and improve power efficiency, the board design avoids direct 12 V conversion stages and instead implements a localized 5 V to 3.3 V regulation stage.

Several LDO regulators were evaluated, including devices from **OnSemi 8705**(best but DFN package make it harder to solder) and Texas Instruments TPS&TLV series. Considering package availability, solderability, and procurement ease, the **TLV75733PDBR LDO regulator was selected**.

4. Power Consumption Estimation

STM32F407 microcontroller – approximately 240 mA (worst-case operating conditions)

TJA1051 CAN transceiver – approximately 110 mA

NTC sensing network – approximately 4.29 mA

The estimated total current consumption of the module, including engineering design margin for transient loads and peripheral components, is approximately **310 mA**. So 1A regulator was chosen

Key Component Features (Summary)

Key features derived from respective datasheets include:

STM32F407VETx:

- ARM Cortex-M4 core architecture
- 100pins LQFP pavkage
- Integrated multi-channel ADC subsystem
- Rich peripheral ecosystem and development support

TJA1051T/3 CAN Transceiver:

- 1Mbps High-speed CAN transceiver compliant with ISO CAN standards
- Robust differential signaling for noise immunity
- Suitable for automotive and industrial CAN networks

TLV75733 LDO Regulator:

- 3.3 V regulated output
- Low dropout voltage
- High PSRR (~46 dB at 100 kHz) improving supply noise rejection
- Compact package suitable for PCB integration

standard quartz crystal oscillator footprint has been incorporated on the PCB. The selection of a common industry-standard crystal package was primarily driven by ease of assembly and soldering during prototyping and small-batch manufacturing.

Although more advanced timing solutions such as Temperature Compensated Crystal Oscillators (TCXO) and MEMS oscillators offer improved frequency stability and temperature performance, these devices were not considered necessary for the current stage of development.

Instead, the design maintains hardware flexibility, allowing the oscillator value and type to be finalized later based on system validation results and timing requirements of the microcontroller and communication interfaces.

Design Calculations

Note: All calculations are theoretical, specially CAN and ADC timing

Temperature Resolution Calculation

Temperature measurements are obtained using **a 10 kΩ MF55 NTC thermistor** connected in a voltage divider configuration with a 10 kΩ reference resistor. The divider output voltage is measured by the ADC of the STM32F407VET6 microcontroller.

The ADC reference voltage is **3.3 V**.

The ADC of the STM32F407 has 12-bit resolution.

Number of ADC levels:

$2^{12} = 4096$

The voltage resolution of the ADC is therefore:

$V_{LSB} = V_{ref} / 2^{12}$

$V_{LSB} = 3.3\ V / 4096$

$V_{LSB} = 0.000805\ V$

$V_{LSB} \approx 0.805\ mV$

Thus, **the smallest measurable voltage change by the ADC is 0.805 mV per count**.

Voltage Divider Sensitivity at 25°C

At 25°C, the thermistor resistance is:

$R_{NTC} = 10\ k\Omega$

The divider consists of:

R1 = NTC thermistor
R2 = 10 kΩ reference resistor
Vcc = 3.3 V

The output voltage at 25°C is:

$V_{out} = Vcc \times (R2 / (R1 + R2))$

$V_{out} = 3.3 \times (10k / (10k + 10k))$

$V_{out} = 3.3 \times 0.5$

V_out = 1.65 V

For a typical 10 kΩ NTC thermistor with β ≈ 3950, the voltage sensitivity around 25°C is approximately:

dV/dT ≈ 36 mV per °C

This means the output voltage changes by approximately 36 mV for every 1°C temperature change.

Temperature resolution can be calculated as:

Temperature Resolution = ADC Voltage Resolution / Voltage Sensitivity

Temperature Resolution = 0.805 mV / 36 mV per °C

Temperature Resolution ≈ **0.022 °C**

Final Result

The theoretical temperature resolution of the system is approximately:

0.02 °C per ADC count

CAN Bus Bandwidth Calculation

The temperature extension modules communicate with the master controller using the CAN 2.0A protocol. The purpose of this calculation is to estimate the bandwidth required to transmit all temperature data from the distributed ADC boards to the master BMS controller.

The system architecture consists of six temperature acquisition boards, each performing local ADC conversion and transmitting the measured data to the master node.

Temperature Data Size per Board

Each board samples **13 ADC channels**.

The ADC resolution of the STM32 microcontroller is 12 bits.

Total data generated per board:

Total bits per board= Number of channels × ADC resolution

Total bits per board= 13 × 12

Total bits per board= 156 bits

Converting to bytes:

156 / 8 = 19.5 bytes

For transmission purposes this is rounded to:**20 bytes per board**

CAN Frame Payload Requirement

The CAN 2.0A standard supports a maximum payload of 8 bytes per frame.

Frames per board= Total data bytes / Payload per frame

Frames per board= 20 / 8

Frames per board ≈ 2.5

Therefore, the value is rounded up to:3 CAN frames per board

Total Frames in the System

The system contains 6 temperature acquisition boards.

Total CAN frames required:

Total Frames= Boards × Frames per board

Total Frames= 6 × 3

Total Frames= 18 CAN frames

A standard CAN 2.0A frame has 8 bytes data space.

Including additional control and inter-frame spacing, the effective frame size is approximately:

111 bits (without bit stuffing) Total Data Transmission Size

Total bits transmitted for all temperature boards:

Total bits= Frames × Bits per frame

Total bits= 18 × 111

Total bits= 1998 bits(TOTAL DATA LOAD)

CAN Bus Transmission Time

The CAN network operates at a bus speed of:**500 kbps**

Time required to transmit the total data:

Transmission time= Total bits / Bus speed

Transmission time= 1998 / 500000≈ 0.003996 seconds ≈ 4 ms

Maximum Scan Rate

The total time required to transmit temperature data from all six boards is approximately 4 ms.

Therefore, the theoretical maximum temperature scan rate is:

Scan rate= 1 / Transmission time

Scan rate= 1 / 0.004

Scan rate ≈ 250 scans per second

Result

The CAN bus bandwidth is sufficient to support the temperature extension system, allowing **approximately 250 full system scans per second**.

This is significantly higher than the system requirement of one complete acquisition cycle per second, ensuring reliable and timely temperature monitoring within the battery management system.

ADC Sampling Time Calculation

The temperature extension module performs analog-to-digital conversion using the internal ADC of the STM32F407 microcontroller. The objective of this calculation is to estimate the time required to sample all temperature channels on a single board.

The ADC in the STM32F407 is clocked from the **APB2 peripheral bus**, which operates at a **maximum frequency of 84 MHz**.

ADC Clock Configuration

The ADC clock is derived from the APB2 clock through an internal prescaler.

prescaler of 4 is selected.

ADC Clock = APB2 Clock / Prescaler

ADC Clock = 84 MHz / 4

ADC Clock = 21 MHz

ADC Conversion Time

According to the STM32F407 datasheet, the total conversion time is given by:

ADC Conversion Time = Sampling Time + Conversion Cycles

Where the conversion cycles for a 12-bit ADC are:

Conversion Cycles = 12.5 cycles

For stable measurements with high source impedance sensors such as NTC thermistors, a sampling time of 84 cycles is selected.

Total Cycles = Sampling Cycles + Conversion Cycles

Total Cycles = 84 + 12.5

Total Cycles = 96.5 cycles

The time required for one conversion is:

Conversion Time = Total Cycles / ADC Clock

Conversion Time = 96.5 / 21 MHz

Conversion Time ≈ 4.59 µs

Each temperature board samples 13 ADC channels.

Total sampling time per board:

Sampling Time per Board= Channels × Conversion Time

Sampling Time per Board= 13 × 4.59 µs

Sampling Time per Board ≈ 59.7 µs

Sampling Time per Board ≈ 60 µs

The total time required to sample all 13 temperature channels on one board is approximately: 60 µs

This conversion time is significantly smaller than the CAN transmission time (~4 ms) required to send the data over the communication network. Therefore, the ADC sampling process does not limit the system performance.

The system can comfortably support the required temperature acquisition rate while maintaining sufficient processing time for filtering and communication tasks.

Power Consumption and LDO Voltage Drop Analysis

The temperature extension board is powered from the 5 V GLV supply rail. The onboard digital circuitry operates at 3.3 V, which is generated using the TLV75733 low dropout regulator (LDO). The CAN transceiver operates directly from the 5 V rail.

The total power consumption is determined by the following components:

- Microcontroller
- CAN transceiver

- NTC sensing network
- LDO quiescent current
- LDO power dissipation
- Microcontroller Power Consumption

The STM32F407 microcontroller operates at 3.3 V.

Maximum current consumption (datasheet estimate):

I_MCU ≈ 240 mA

Power consumption:

P_MCU = V × I

P_MCU = 3.3 × 0.24

P_MCU = 0.792 W

CAN Transceiver Power Consumption

The CAN communication interface uses the TJA1051 CAN transceiver, which operates from the 5 V rail.

Typical operating current:

I_CAN ≈ 110 mA

P_CAN = V × I

P_CAN = 5 × 0.11

P_CAN = 0.55 W

NTC Voltage Divider Power

Each temperature channel uses a 10 kΩ NTC thermistor and 10 kΩ reference resistor forming a voltage divider.

Total resistance:

R_total = 10 kΩ + 10 kΩ

R_total = 20 kΩ

Divider current:

I_div = V / R

I_div = 3.3 / 20000

I_div = 0.165 mA

Each board measures 13 channels.

Total NTC current:

I_NTC = 13 × 0.165

I_NTC ≈ 2.15 mA

P_NTC = V × I

P_NTC = 3.3 × 0.00215

P_NTC ≈ 7.1 mW

The TLV75733 has a typical quiescent current of approximately:

I_q ≈ 25 µA

Power consumption due to quiescent current:

P_q = V_in × I_q

P_q = 5 × 25 µA

P_q ≈ 0.125 mW

This value is negligible compared to the rest of the system.

Total Current on 3.3 V Rail

Total load current supplied by the regulator:

I_load = I_MCU + I_NTC

I_load = 240 mA + 2.15 mA

I_load ≈ 242 mA

The selected TLV75733 regulator has a typical **dropout voltage of approximately 120 mV at 500 mA** load current, which is significantly lower than the available 1.7 V headroom.

Therefore, the regulator operates well within its regulation region.

Power dissipation in the LDO:

P_LDO = V_drop × I_load

P_LDO = 1.7 × 0.242

P_LDO ≈ 0.41 W

Total power consumption per board:

P_total = P_MCU + P_CAN + P_NTC + P_LDO

P_total = 0.792 + 0.55 + 0.007 + 0.41

P_total ≈ 1.76 W

The complete system contains 6 temperature extension boards.

Total system power:

P_system = 6 × 1.76

P_system ≈ 10.56 W

Total current drawn from the 5 V GLV supply:

I_system = P_system / V

I_system = 10.56 / 5

I_system ≈ 2.11 A

Each temperature extension board consumes approximately:1.76 W

The complete system consisting of six boards consumes approximately:10.5 W from the 5 V supply rail.

Key Results

PARAMETER	RESULT
TEMPERATURE RESOLUTION	**0.02 °C per ADC count**
ADC SAMPLING TIME (13 CHANNELS)	**≈ 60 µs**
CAN TRANSMISSION TIME (6 BOARDS)	**≈ 4 ms**
MAXIMUM SYSTEM SCAN RATE	**≈ 250 scans/sec**
POWER CONSUMPTION PER BOARD	**≈ 1.76 W**
TOTAL POWER FOR 6 BOARDS	**≈ 10.5 W**

Design Validation Summary

- The proposed temperature extension module was evaluated through analytical calculations to verify that it meets the design goals of accurate temperature measurement, reliable communication, and efficient power consumption.
- The temperature measurement system, based on 10 kΩ MF55 NTC thermistors and the 12-bit ADC of the STM32F407, provides a theoretical resolution of approximately **0.02°C per ADC count**. Considering thermistor tolerance, β variation, and resistor tolerance, the overall worst-case measurement accuracy is **approximately ±2°C**, which is acceptable for battery thermal monitoring applications.
- The ADC sampling performance was calculated to require approximately 60 µs to sample all 13 channels on a single board. This ensures that the sensing subsystem does not limit system performance.
- For communication, each board transmits its temperature data using 3 CAN frames, resulting in a total of 18 frames for the six-board system. At a CAN bus speed of 500 kbps, the total transmission time is approximately 4 ms, allowing a theoretical scan rate of **about 250 scans per second**, which exceeds the system requirement.
- The power consumption of a single board is approximately 1.76 W, including microcontroller operation, CAN transceiver power, and LDO dissipation. The complete system of six boards consumes **approximately 10.5 W**, which is within the capacity of the GLV supply system.
- Additionally, the thermistor self-heating analysis shows a temperature rise of approximately **0.27°C**, which is negligible compared to the overall measurement uncertainty.
- Overall, the analysis confirms that the proposed architecture satisfies the system requirements and provides a reliable and scalable solution for extending temperature sensing capability in the battery management system.

PCB Design Considerations

The PCB layout of the temperature extension module was organized into four functional blocks to simplify routing, reduce noise coupling, and improve maintainability. The major blocks are:

- **Power Supply Block**
- **NTC Sensing Block**
- **CAN Communication Block**
- **Microcontroller Block**

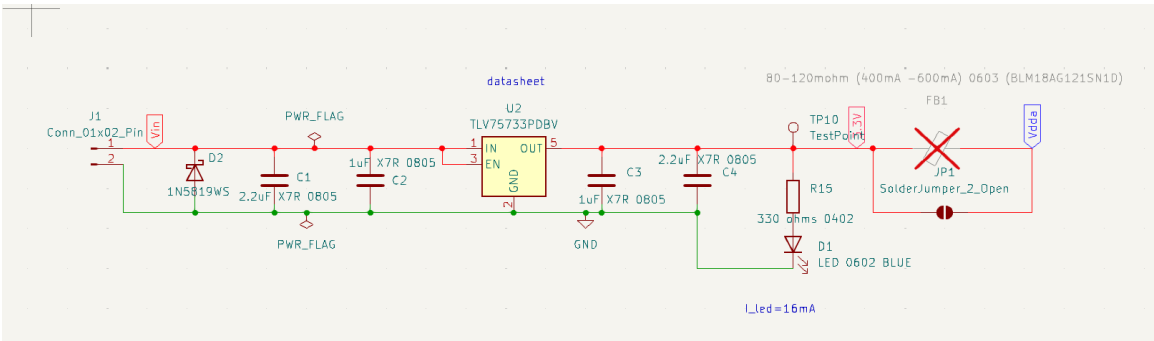
This modular layout ensures that sensitive analog circuitry is isolated from digital switching noise while maintaining efficient routing across the board.

1. Power Supply Block

At the power input terminals, a 5819 Schottky diode is used to provide reverse polarity protection for the board.

- Input filtering for the TLV75733PDBR LDO regulator is implemented using 2.2 µF and 1 µF capacitors, placed close to the regulator input pins. At the regulator output, 1 µF and 2.2 µF capacitors are used to stabilize the output voltage and ensure proper regulator operation as recommended in the datasheet.
- A power indication LED is also included on the board to provide visual confirmation of power availability.
- To improve ADC measurement stability, the analog and digital supply rails are separated. A ferrite bead is used to isolate the analog supply (VDDA) from the main digital supply. The VDDA rail powers the following components:
  MCU ADC reference
  VREF circuitry

The ferrite bead is included as an optional component. In cases where the ADC environment is sufficiently clean or if cost optimization is required, the ferrite bead can be bypassed. Therefore, the ferrite bead is marked as **DNP** (Do Not Populate) and a solder jumper is provided to allow direct connection if needed.

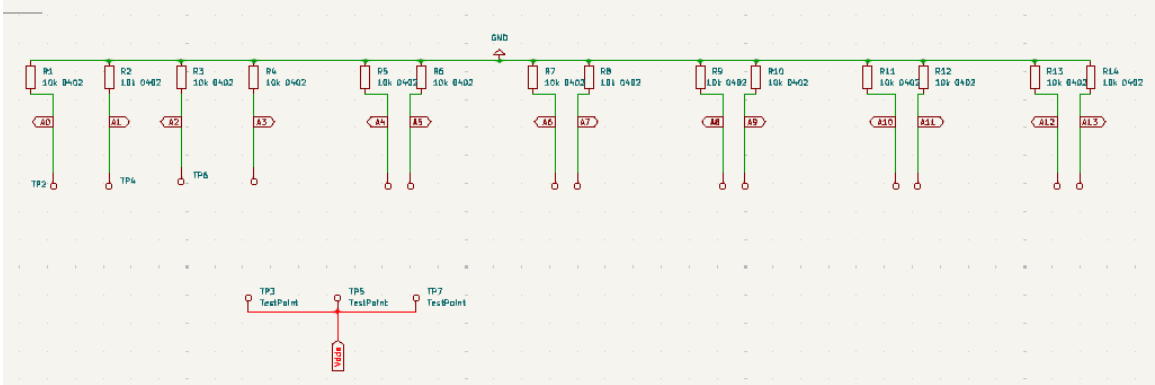


2. NTC Sensing Block

Since this board revision is intended as a testing and validation version, only three positive nodes are currently provided for thermistor connections.

Instead of using permanent connectors, test points are provided to allow flexibility during initial testing and validation. NTC thermistors can be soldered directly onto these test points.

This approach provides sufficient accessibility during the prototype stage while leaving room for connector-based implementation in future revisions.



3. CAN Communication Block

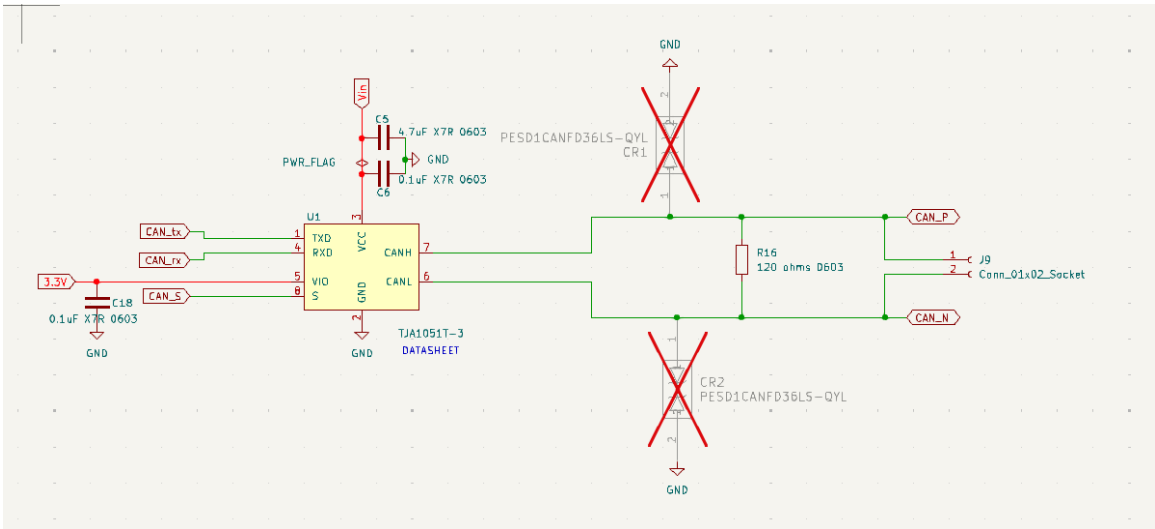
- The CAN communication interface follows the recommended application circuit provided in the transceiver datasheet.
- A CAN Silent Mode pin is made accessible to the microcontroller. This allows future firmware control of the CAN transceiver to:
  disable transmission during arbitration errors
  place the transceiver in low power mode

The **CANH and CANL differential pair** are routed together to maintain signal integrity.

A 120 Ω termination resistor is used for bus termination. Some CAN designs implement a split termination (60 Ω + 60 Ω with a capacitor) for improved common-mode noise suppression. However, since the expected wiring harness length is short, a standard 120 Ω termination is considered sufficient for this design.

Similarly, **common mode chokes are not used because the communication distance is short** and the electromagnetic environment is manageable.

Optional ESD protection are included in the design as DNP components, allowing them to be populated later if additional protection is required.



4. Microcontroller Block

- All decoupling and filtering capacitors are implemented as recommended in the microcontroller datasheet.
- The **VCAP1 and VCAP2 pins are connected to 2.2 µF** capacitors to stabilize the internal voltage regulator of the MCU.
  Important firmware note:
  The internal regulator must remain enabled, as disabling it may affect correct MCU operation.
- **Both BOOT0 and BOOT1** pins are made externally accessible to allow flexible boot configuration during development and debugging. Boot configuration resistors are placed according to the datasheet recommendations.
- A **reset pushbutton** is included on the board to allow manual reset of the microcontroller. The reset line is also accessible through the programming interface to support firmware flashing.
- For programming and debugging, the **SWD interface is provided**:
  SWDIO
  SWCLK
- Additionally, **SWO is exposed to support advanced debugging features when using GCC-based debugging tools.**

Important Note:
All hardware design decisions related to the microcontroller connections, decoupling, and boot configuration should be verified with the official STM32F407 datasheet and reference manual.

PCB Stack-up

The PCB for the temperature extension module follows the **6-layer stack-up** specification provided **by Lion Circuits**. A multi-layer board was selected to improve signal integrity, provide a solid ground reference, and simplify routing of power, analog signals, and CAN communication traces.

The stack-up structure is shown below.

Signal

GND

Signal

PWR

GND

Signal

Layer	Material type	Thickness	
Top Layer1	Copper	0.035 mm	
Prepreg	2116*1	0.12 mm	
Inner Layer2	Copper	0.035 mm	
Core	NP-140	0.36 mm	0.4 mm (with copper core)
Inner Layer3	Copper	0.035 mm	
Prepreg	2116*4	0.48 mm	
Inner Layer4	Copper	0.035 mm	
Core	NP-140	0.36 mm	0.4 mm (with copper core)
Inner Layer5	Copper	0.035 mm	
Prepreg	2116*1	0.12 mm	
Bottom Layer6	Copper	0.035 mm	

Stack-up Considerations

The multi-layer stack-up provides several advantages for the design:

Dedicated inner layers allow the implementation of solid ground and power planes, improving noise immunity.

The stack-up helps maintain controlled impedance routing for CAN communication traces.

Internal layers reduce electromagnetic interference (EMI) and improve overall signal integrity.

Sensitive analog traces related to the NTC sensing network and ADC inputs can be shielded by adjacent ground planes.

Test points are provided to evaluate, test and Debug

Since the PCB does not have any High-speed signal, impedance-controlled PCB will not be necessary.

Using a 6-layer PCB also simplifies routing density and improves reliability compared to a 2-layer or 4-layer board, particularly in mixed-signal designs containing both analog sensing circuits and high-speed digital communication.

Firmware Development and Evaluation Recommendations

- Strongly **recommend Interrupt based scheduling** as CAN protocol Latency is much higher than ADC, gives a lot of headroom for Temperature conversions and eliminates risk of CAN bus being pulled down.
- Suggested use DMA based continuous sampling
- Use CAN error bit and Watchdog extensively
- Dynamic priority allotment to all 6 boards may decrease performance kindly check

During firmware development and validation of the temperature extension module, the following practices are recommended to evaluate system performance and reliability.

1. **CAN Arbitration Evaluation**
   Test CAN communication with multiple nodes transmitting simultaneously to verify correct arbitration behaviour and ensure no persistent arbitration errors occur.

2. **Operating System Evaluation**
   Initial development may be performed using **bare-metal firmware or FreeRTOS** to evaluate timing and task behaviour. For the final system, migration to the **Zephyr RTOS** is recommended on both the master controller and temperature support modules.

3. **ADC and CAN Operation Modes**
   Evaluate different operating modes of the ADC and CAN peripherals, including **polling, interrupts, and DMA-based operation**. This helps determine the most efficient configuration for **sampling performance and communication latency**.

4. **Timing and Performance Analysis**
   Integrate firmware with timing analysis tools such as **Segger System View** to monitor **task execution time, interrupt latency, and overall system timing behaviour**.

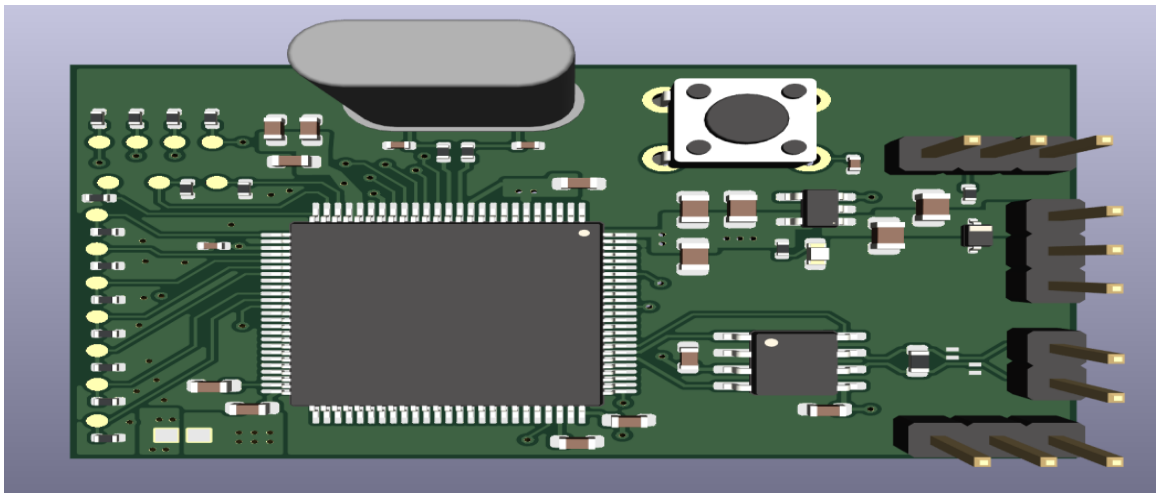
5. **Hardware Signal Analysis**
   Use available laboratory instruments such as a digital logic analyser, oscilloscope, and vector analyser to **observe communication signals, verify timing behaviour, and evaluate signal integrity** during system testing.

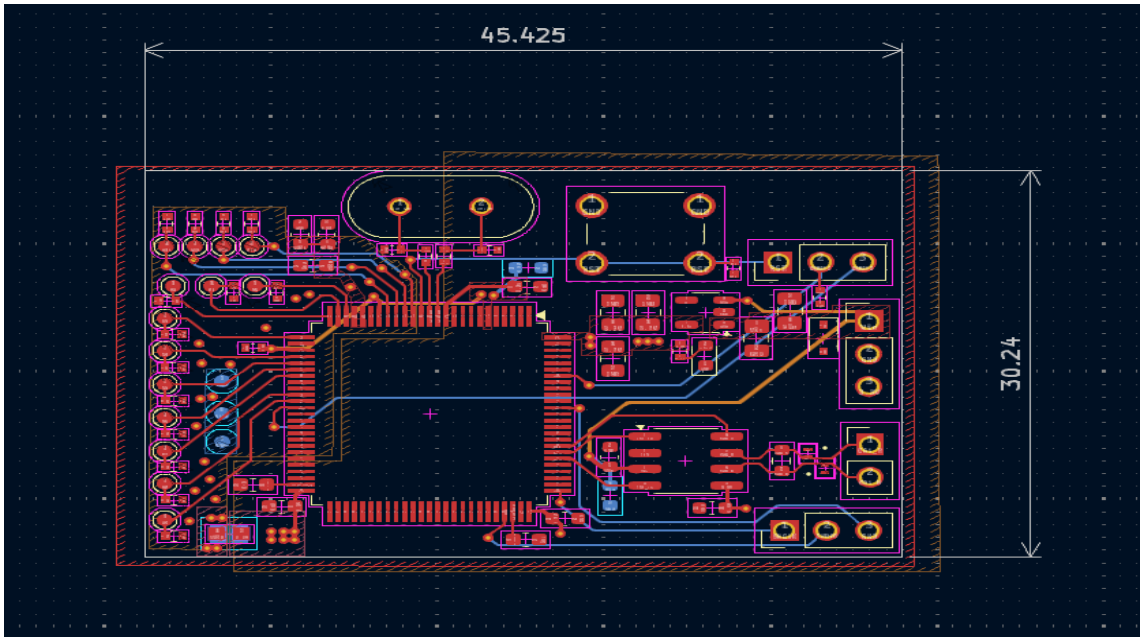
   During final assembly Kindly use equal length twisted pair wires

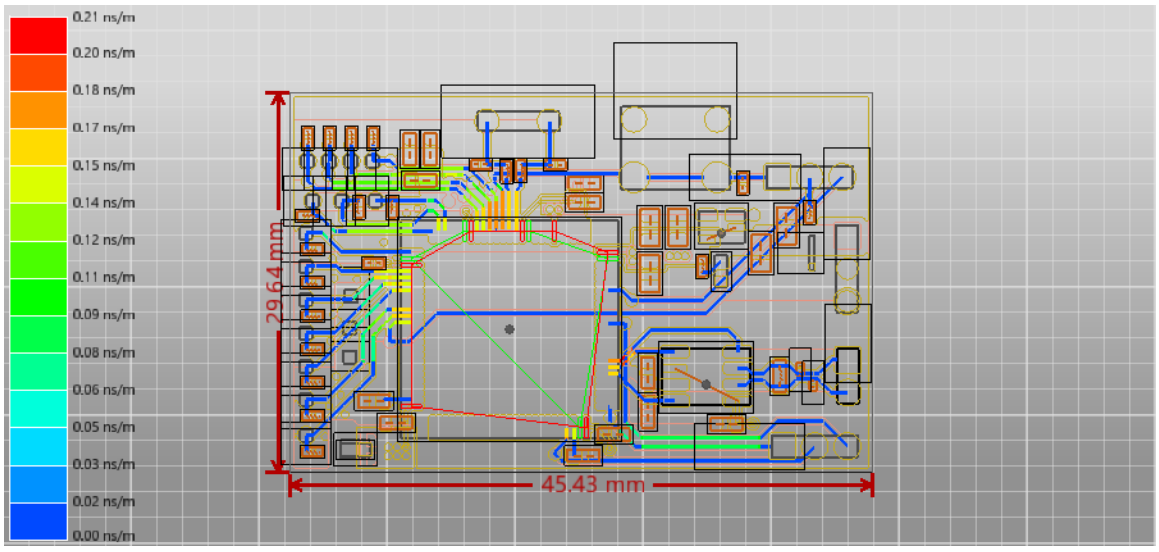
BOM

Rs.1296/- for 2 boards according to JLC PCB

Designator	Footprint	Quantity	Value
C1, C4	0805	2	2.2µF X7R 0805
C10, C11	0603	2	2.2µF X7R 0603
C12	0603	1	1µF X7R 0603
C13, C14, C15, C16, C18, C6, C8, C9	0603	8	0.1µF X7R 0603
C17	0402	1	0.1µF 0402
C19, C20	0402	2	10pF 0402
C2, C3	0805	2	1µF X7R 0805
C21, C5	0603	2	4.7µF X7R 0603
C7	0805	1	4.7µF X7R 0805
D1	0603	1	LED Blue
D2	D_SOD-323	1	1N5819WS
J1,J9	2.54mm_Vertical	2	Conn_01x02_Pin
J10, J11	2.54mm_Vertical	2	Conn_01x03_Socket
R1, R10, R11, R12, R13, R14, R17, R2, R3, R4, R5, R6, R7, R8, R9, R20	0402	16	10kΩ 0402
R15	0402	1	330Ω 0402
R16	0603	1	120Ω 0603
R18, R19	0402	2	33Ω 0402
R21	0402	1	510Ω 0402
SW2	SW_PUSH_6mm	1	Push Button Switch
U1	SOIC-8_3.9x4.9mm_P1.27mm	1	TJA1051T-3
U2	SOT-23-5	1	TLV75733PDBV
U3	LQFP-100_14x14mm_P0.5mm	1	STM32F407VETx
Y1	Crystal_HC49-U_Vertical	1	26 MHz Crystal (YXC HC-49US)




45.425
30.24


0.21 ns/m
0.20 ns/m
0.18 ns/m
0.17 ns/m
0.15 ns/m
0.14 ns/m
0.12 ns/m
0.11 ns/m
0.09 ns/m
0.08 ns/m
0.06 ns/m
0.05 ns/m
0.03 ns/m
0.02 ns/m
0.00 ns/m
29.64 mm
45.43 mm